Implementation work specification

Measurement package for *Layer-aware post-quantum cryptography for blockchain-based energy trading*

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[DATE]

1 Purpose and scope

This document specifies the implementation and measurement work required for a journal submission to *Computers & Electrical Engineering*. The analytical model, the protocol design and the manuscript are already written. What is missing is the empirical half: a working testbed and a set of measurements that either confirm or refute the model.

You are not being asked to reproduce a predicted result. You are being asked to measure what the system actually does. If the measurements contradict the model, that is a finding, not a failure, and it must be reported as measured. Do not tune an experiment until it matches an expected curve.

Explicitly out of scope. Power-flow simulation, voltage or thermal analysis, market clearing, and any claim about the physical electricity network. The manuscript treats physical feasibility as an external constraint. Do not model it, and do not describe emulated network links as power lines.

2 Environment

All versions **MUST** be pinned and recorded in `meta.json` (Sec. 6). "Latest" is not a version.

Component	Requirement	Notes
Blockchain	Hyperledger Fabric 2.5.x LTS	Raft ordering. **MUST** not use Solo.
PQC library	`liboqs` + Go bindings	pin the exact release tag
Load driver	Hyperledger Caliper	community-standard harness
Net emulation	Mininet	communication network only
Server host	x86_64, $\geq$4 cores	record CPU, clock, RAM, OS
Meter host	single-board computer	record SoC, clock, RAM, OS
Power meter	USB meter or INA219	for E8 only

Parameter sets. `ML-KEM-768`, `ML-DSA-44`, `ML-DSA-65`, `ML-DSA-87`, `Falcon-512`, `SPHINCS+-SHA2-128s`, and `ECDSA P-256` as the classical baseline.

Two traps that have caught others.

1. `SPHINCS+-128s` and `SPHINCS+-128f` differ by more than an order of magnitude in signing time. We need the `s` (small-signature) variant. Benchmarking `f` by accident is the single most common error in this area.
2. Falcon in current `liboqs` tracks the round-3 submission, not the final FIPS 206 draft. Record this in `meta.json`; the manuscript already states the caveat.

3 Measurement discipline

These rules apply to every experiment and are not negotiable, because the manuscript states them in its methodology section.

- $\geq 10^3$ iterations per data point; discard a warm-up phase.
- Report **median, p95 and p99** — never the mean alone. The penalty mechanism is governed by worst case, not average.
- Pin the process to a fixed core set and disable frequency scaling for timing runs. Record the governor setting.
- Never run a timing measurement inside Mininet while other emulated hosts are doing cryptographic work. CPU contention on a shared kernel will corrupt the result. Measure crypto (E0) and network (E9) separately and compose them.
- Record raw samples, not just summaries. Summaries can be recomputed; samples cannot be recovered.

4 The sanity gate

Before producing any plot, run your E0 numbers through the checker that ships with the plotting script:

```
python3 -c "import make_figures as m; \
  m.check({'ML-DSA-65': (0.21, 0.08), 'ECDSA P-256': (0.04, 0.11)})"
```

It compares each measurement against a plausible band derived from published benchmarks and flags anything outside it. A flag does *not* mean your measurement is wrong — it means either the measurement or the band needs an explanation before it goes near the manuscript. In a previous draft in this group, an `ECDSA` signing time was recorded that this gate would have flagged as 333× out of band. Run it every time.

Definition of done: Every E0 value either falls inside its band, or has a written one-line explanation of why it does not.

5 Deliverables

Each experiment produces one CSV with the exact schema below. The plotting script reads these directly; if the schema matches, no rework is needed. Put them in `data/`.

E0 — primitive microbenchmarks

File: `data/e0_primitives.csv`

```
platform,scheme,operation,n_iter,median_ms,p95_ms,p99_ms,stddev_ms
server,ML-DSA-65,keygen,10000,...,...,...,...
server,ML-DSA-65,sign,10000,...,...,...,...
server,ML-DSA-65,verify,10000,...,...,...,...
meter,ML-DSA-65,sign,1000,...,...,...,...
```

`platform` $\in$ {server, meter}; `operation` $\in$ {keygen, sign, verify, encaps, decaps}. All seven schemes, both platforms.

Definition of done: CSV complete for all scheme × platform × operation combinations; sanity gate passed or exceptions documented.

E1 — Layer-1 with post-quantum identities

File: `data/e1_fabric.csv`

```
config,identity_bytes,endorse_median_ms,endorse_p95_ms,
commit_median_ms,tps_sustained,block_utilisation
ECDSA,...,...,...,...,...,...
ML-DSA-44,...,...,...,...,...,...
ML-DSA-65,...,...,...,...,...,...
SLH-DSA,...,...,...,...,...,...
```

`block_utilisation` is the fraction of block capacity consumed by ordinary traffic. This is the parameter ρ in the model and is the one number here that feeds the central result — measure it carefully.

Definition of done: Four configurations measured under an identical Caliper workload profile; the profile file is committed to the repository.

E2 — channel state machine

File: `data/e2_statemachine.csv`

```
config,transition,message_bytes,rtt_median_ms,rtt_p95_ms
classical,commitment_update,...,...,...
uniform_mldsa,commitment_update,...,...,...
layer_aware,commitment_update,...,...,...
```

`transition` ∈ {channel_setup, commitment_update, htlc_add, htlc_settle, funding, coop_close, force_close, penalty}. `config` ∈ {classical, uniform_mldsa, layer_aware}.

The `penalty` row is the most important line in this entire package: its `message_bytes` is the quantity the central result depends on. Measure the real serialised transaction, do not compute it.

Definition of done: All eight transitions, all three configurations, with the penalty transaction measured from a real force-close-and-punish execution.

E5 — watchtower scalability

File: `data/e5_watchtower.csv`

```
config,n_states,blob_bytes_median,total_bytes,scan_cpu_ms
layer_aware,1000,...,...,...
layer_aware,10000,...,...,...
```

Sweep `n_states` over at least four decades so the growth trend is visible. Record both storage and scan CPU: the claim under test is that the binding constraint moves from CPU to storage.

Definition of done: Growth is measured, not extrapolated from a single point.

E7 — end-to-end prosumer workload

File: `data/e7_endtoend.csv` (raw samples, for a CDF)

```
config,run_id,sample_idx,latency_ms
layer_aware,1,0,...
```

`config` ∈ {l1_only, sets_reimpl, classical_l2, layer_aware_l2}. Drive with a real photovoltaic and demand trace at a 15-minute settlement interval; record the dataset name and licence in `meta.json`.

The `sets_reimpl` configuration is *our reimplementation* of a published design. Label it that way in every artefact you produce. It is not a measurement of the original authors' system and **MUST** never be described as one.

Definition of done: $\geq 10^3$ samples per configuration; trace source documented.

E8 — energy on constrained hardware

File: `data/e8_energy.csv`

```
scheme,operation,time_ms,idle_w,load_w,energy_mj,n_iter
```

Measure idle power first, then under load; energy per operation is (load − idle) × time. Report the measurement method and the instrument.

Definition of done: Idle baseline recorded separately; instrument identified.

E9 — network sensitivity

File: `data/e9_network.csv`

```
config,rtt_ms,hops,completion_median_ms,completion_p95_ms,n_iter
```

`rtt_ms` $\in \{5, 20, 50, 100\}$; `hops` $\in \{1, \ldots, 5\}$. Use Mininet with `tc netem`. Verify the emulated latency with `ping` before each run and record the verification.

Definition of done: Emulated RTT verified against the configured value; deviation $< 10\%$.

E6 — key-aggregation ablation (optional, high value)

File: `data/e6_aggregation.csv`

```
scheme,aggregated,k_sig,k_pk,tx_bytes
```

Compare a classical aggregated 2-of-2 funding output against two separate post-quantum signatures. This quantifies an open problem the manuscript states; if you have time, do it.

6 Repository and metadata

```
repo/
  README.md how to reproduce, end to end
  meta.json all versions, hardware, dataset provenance
  env/ Fabric config, Caliper profiles, Mininet topos
  src/ harness code
  data/ the CSVs above
  raw/ raw samples, compressed
  figures/ generated PDFs
```

`meta.json` **MUST** record: Fabric version, liboqs release tag and commit, Caliper version, Go version, Mininet version, server and meter hardware, CPU governor, dataset name and licence, and the exact parameter set string for every primitive.

Definition of done: A person with the repository and no access to you can rerun everything from `README.md`.

7 Milestones

Week	Deliverable	Gate
1	Environment up; `meta.json` started	Fabric+Raft runs a trivial chaincode
2	E0 both platforms	sanity gate passed
3	E1 with Caliper	ρ measured
4	E2 including a real penalty transaction	penalty bytes measured
5	E5, E9	trends visible, not extrapolated
6	E7, E8; repository frozen	reproducible from README

Report weekly with: what was measured, what surprised you, what is blocked. A surprising measurement is more valuable than an expected one and should be raised immediately rather than investigated silently.

8 Authorship and credit

Completion of this package to the standard above constitutes a substantial contribution to the submitted manuscript and carries co-authorship, with the CRediT roles *Software*, *Investigation*, *Validation* and *Data curation*. Author order will be agreed before submission. You will be named in the data-availability statement as the author of the archived artefact.

9 Integrity requirements

1. Report measurements as obtained. Never adjust an experiment to approach an expected value.
2. If a result contradicts the model, say so. That is a publishable finding and it is the supervisor's problem to interpret, not yours to hide.
3. Never present a reimplementation of someone else's design as a measurement of their system.
4. Keep raw samples. If a number in the manuscript cannot be traced back to a raw file, it cannot be published.